CHINA · SEMICONDUCTORS · AI

THE BATTLE FOR CHIPS

Computing power, value chains and technological autonomy

Sergio Gevatschnaider



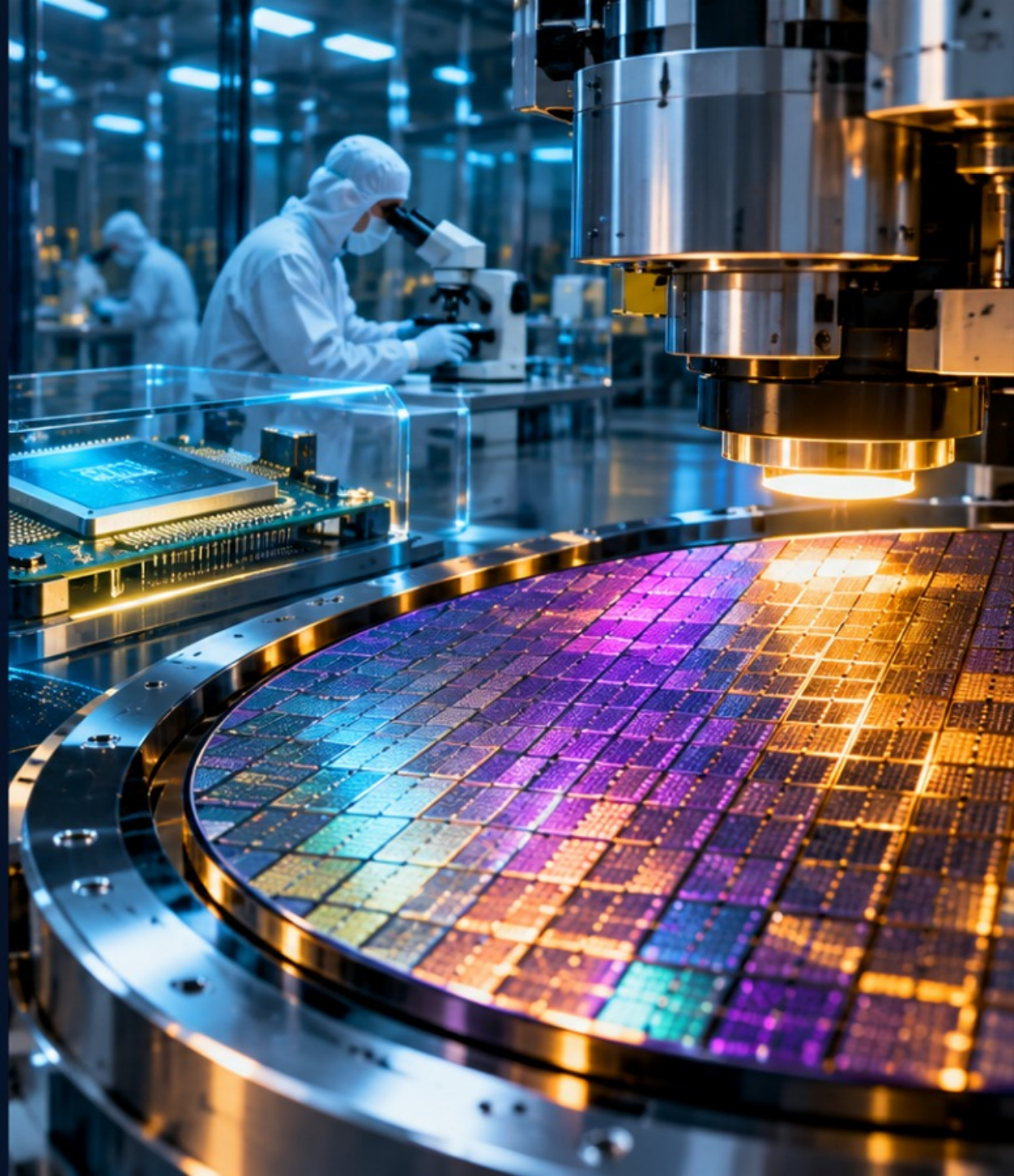
BLOCK 01 / 07

The economics of computing

From models to infrastructure

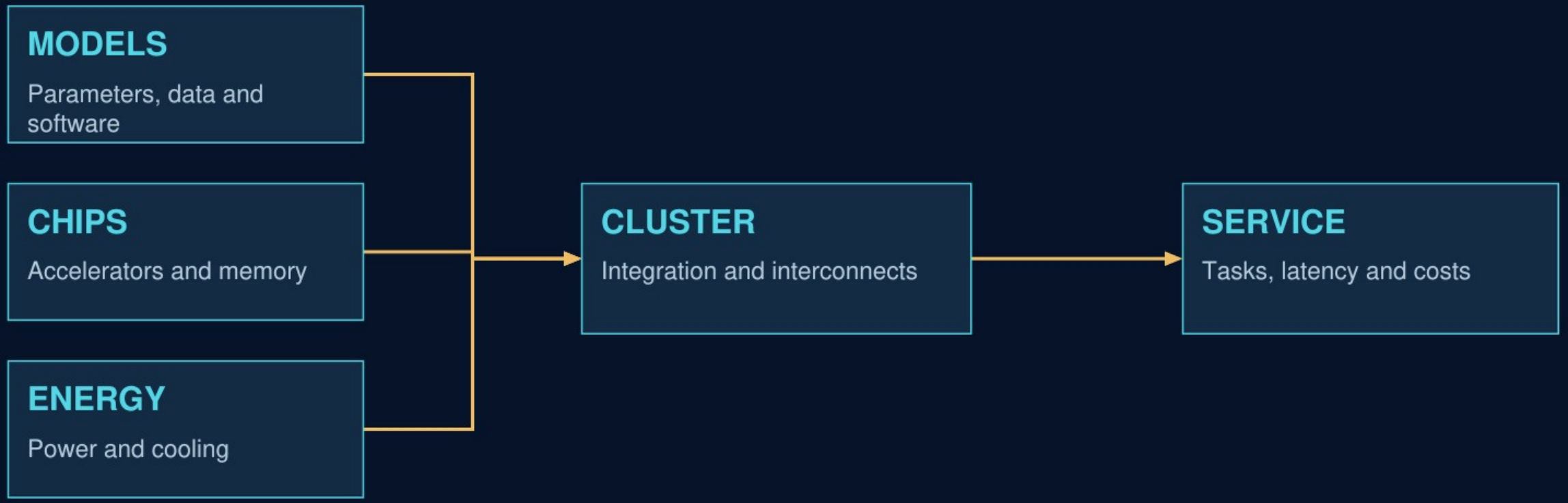
What determines the cost of a unit of AI?

SERGIO GEVATSCHNAIDER



AI depends on interconnected constraints

The algorithmic frontier depends on memory, chips, networks, energy and software.



Dependency relationships: complementary inputs support the service.

Sources in speaker notes · Original schematic diagrams

CPU, GPU and NPU architectures

Compare workloads and programming ecosystems.

01

CPU

General control and latency, with less massive parallelism.

02

GPU

Parallel vector and matrix computation with high flexibility.

03

NPU / ASIC

Specialised acceleration whose efficiency depends on software.

04

HBM

High-bandwidth memory integrated near the accelerator.

Distinguish theoretical FLOPS, usable memory and actual performance.

Sources in speaker notes · Original schematic diagrams

Training and inference

The two phases place different demands on AI infrastructure.

A

TRAINING

Batches, gradients, synchronisation, distributed compute, activation memory and fixed costs.

B

INFERENCE

Prefill, decoding, KV cache, concurrency, response latency and cost per completed task.

An accelerator optimised for training may be less suitable for inference.

Sources in speaker notes · Original schematic diagrams

Bottlenecks can shift

Effective performance depends on the narrowest constraint.



LIMITING STAGE → NETWORK

Teaching indices (0–100), rather than commercial hardware benchmarks.

Sources in speaker notes · Original schematic diagrams

The global semiconductor market

AI expansion increases demand for logic, memory and complete systems.

US$795.6 billion

Worldwide semiconductor sales in 2025, WSTS

13.5 nm

EUV lithography wavelength, ASML

5+

Interdependent strategic segments

These figures describe different quantities: markets, technology and supply chains.

Sources in speaker notes · Original schematic diagrams

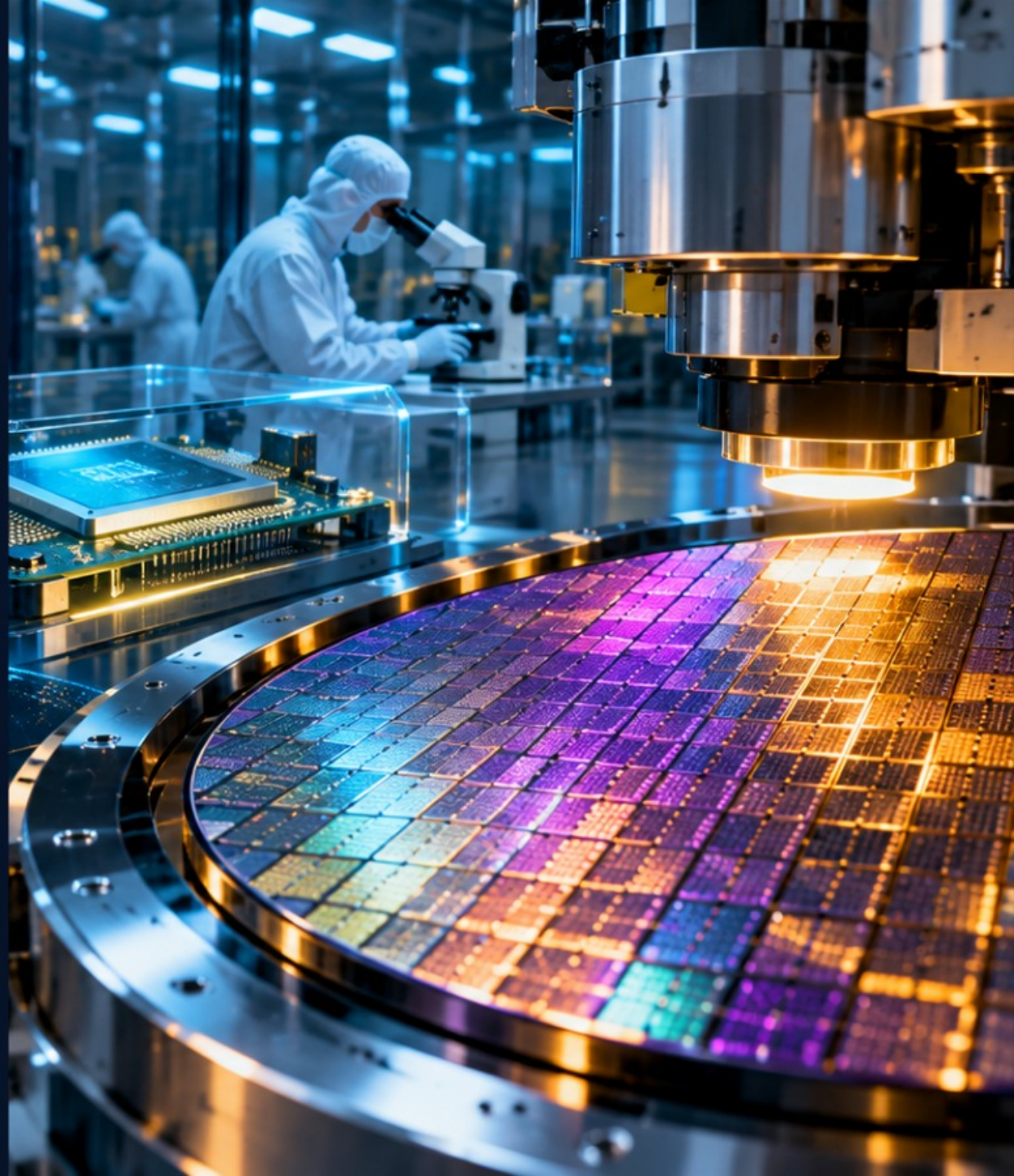
BLOCK 02 / 07

Physics and fabrication

From digital designs to wafers

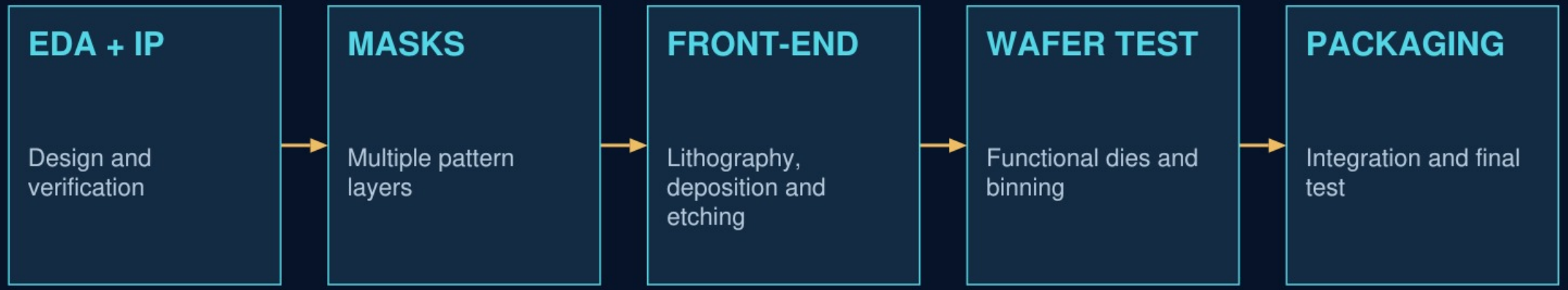
What does an industry need to manufacture advanced chips?

SERGIO GEVATSCHNAIDER



The manufacturing process

Design, masks, processing, testing and packaging require distinct capabilities.



Simplified functional workflow: front-end fabrication repeats many process cycles.

Sources in speaker notes · Original schematic diagrams

DUV and EUV lithography

Lithography creates patterns as one step in a longer process.



EUV operates in a vacuum. Metrology and alignment control both processes. Functional schematic.

Sources in speaker notes · Original schematic diagrams

Process nodes and physical dimensions

Commercial node names no longer represent a universal geometric measurement.

01

DENSITY

Transistors per area, depending on design and libraries.

02

POWER

Consumption under a defined workload and thermal dissipation.

03

PERFORMANCE

Frequency, leakage, variability and yield.

04

COST

Masks, processing steps, equipment and volume.

A node name alone does not establish competitiveness.

Sources in speaker notes · Original schematic diagrams

Yield and manufacturing economics

The share of functional dies affects industrial costs.



40 gross dies

30 functional · 10 defective

YIELD = 75%

40 × 0.75 = 30

Good chips = gross dies × yield

The drawn proportion matches the calculation. Teaching example without a commercial node.

Sources in speaker notes · Original schematic diagrams

Advanced packaging

Chiplets, interposers and HBM make the package a complete system.



System performance depends on integrating logic, memory, interconnects and heat management.

Sources in speaker notes · Original schematic diagrams

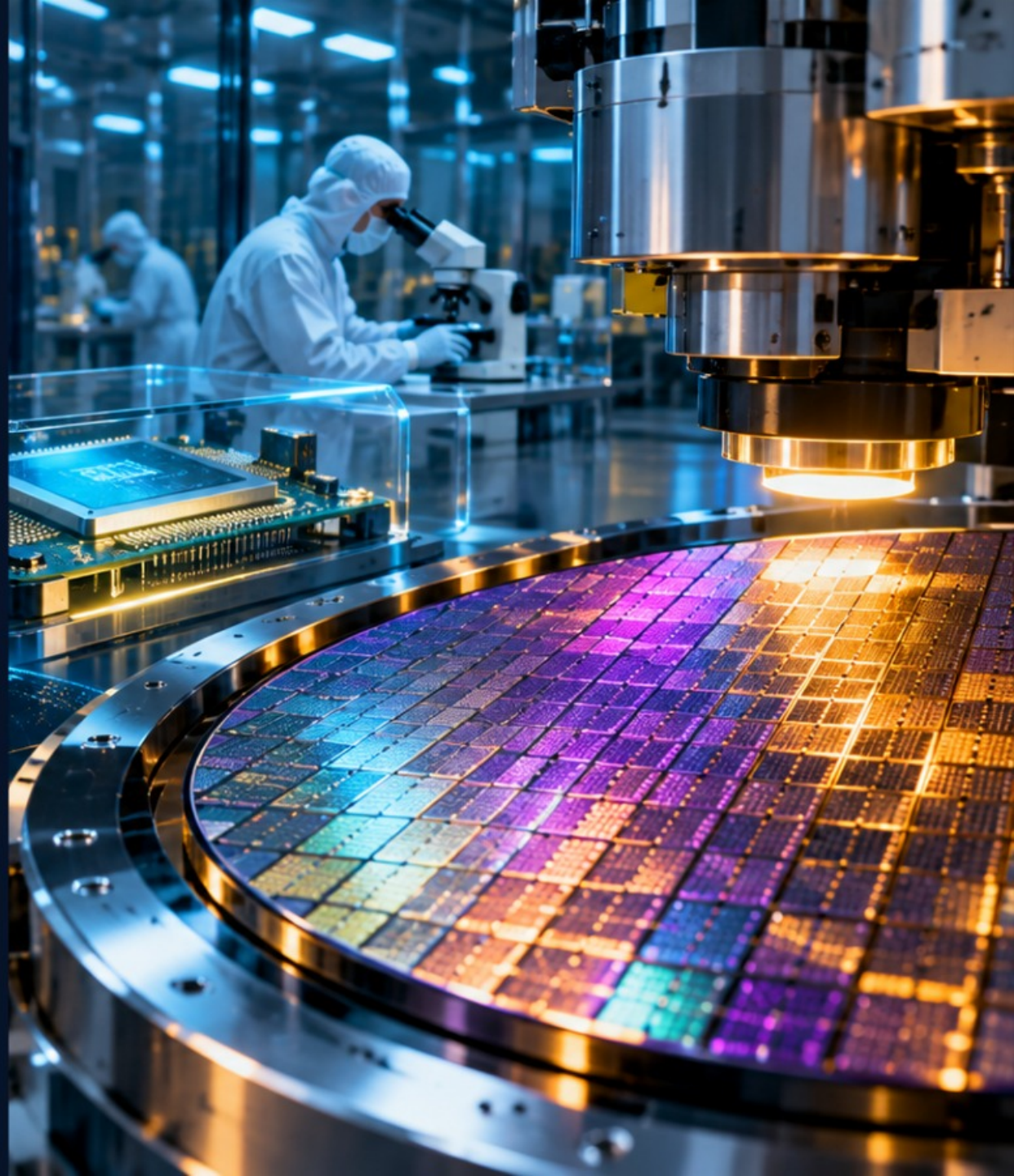
BLOCK 03 / 07

The global semiconductor supply chain

Design, materials, equipment and capacity

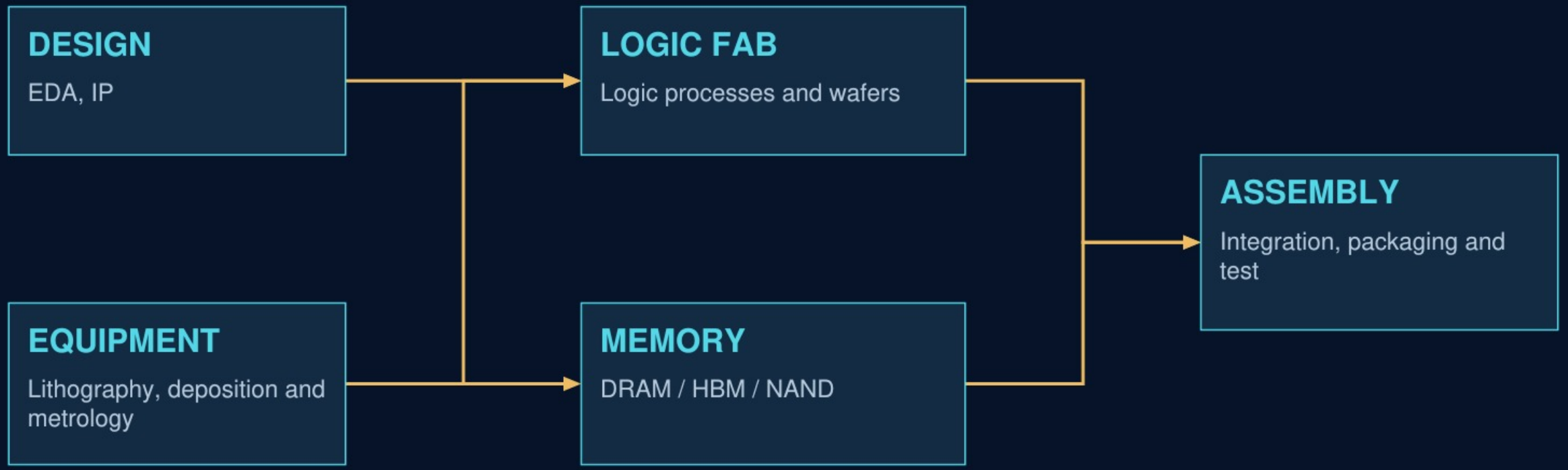
Where is each dependency concentrated?

SERGIO GEVATSCHNAIDER



Global semiconductor architecture

Strategic functions span multiple economies.



Logic and memory are distinct branches that converge in product integration.

Sources in speaker notes · Original schematic diagrams

Dependencies in critical technologies

Substitution proceeds at different speeds for each component.

01

EDA / IP

Design, verification and licensed intellectual property.

02

EUV and metrology

Complex equipment, optics, materials and precision.

03

HBM

Stacked memory, quality control and packaging.

04

Interconnects

Switches, cables, protocols and distributed software.

Assess self-sufficiency by workload and supply-chain stage.

Sources in speaker notes · Original schematic diagrams

Fabrication investment and complexity

A fab combines infrastructure, operations and continuous learning.

01

CAPEX

Buildings, equipment, energy and water treatment.

02

OPEX

Materials, maintenance, technicians and services.

03

LEARNING

Process tuning and systematic defect reduction.

04

TIME

Construction, qualification and production ramp-up.

More nominal capacity may not produce more functional chips.

Sources in speaker notes · Original schematic diagrams

Capacity, yield and demand

Several constraints determine deliveries of completed accelerators.

Equivalent units per month · reference scenario



Output = min(90,000, 80,000, 65,000) = 65,000

HBM limits supply. 65,000 / 120,000 = 54.2% of hypothetical demand.

Sources in speaker notes · Original schematic diagrams

Resilience and concentration

Diversification can reduce risk while increasing costs.

A

EFFICIENCY

Scale, specialisation, learning, availability and low unit costs.

B

RESILIENCE

Redundancy, strategic inventories, substitution, compatibility and alternative suppliers.

Economic strategy balances efficiency, cost and systemic risk.

Sources in speaker notes · Original schematic diagrams

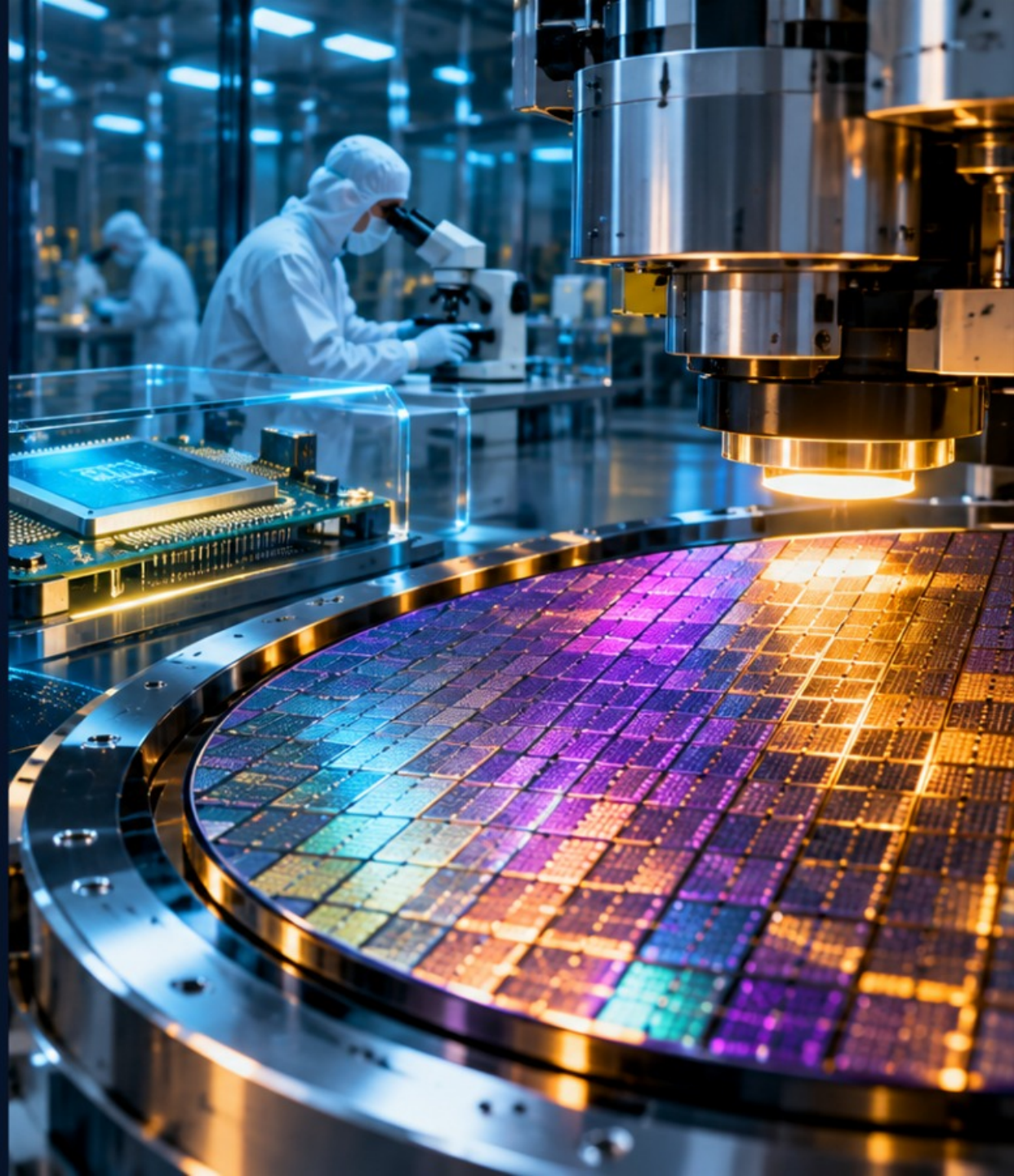
BLOCK 04 / 07

China's semiconductor ecosystem

Firms, specialisations and constraints

Which parts of the stack can China expand locally?

SERGIO GEVATSCHNAIDER



Functions in China's semiconductor ecosystem

Different firms operate in different parts of the supply chain.

01

HUAWEI / HiSilicon

SoC and accelerator design, including the Ascend platform.

02

SMIC

Semiconductor foundry and manufacturing.

03

CAMBRICON

AI accelerator and system design.

04

CXMT / YMTC

Primarily DRAM and NAND memory, respectively.

Distinguish chip design, manufacturing and data-centre integration.

Sources in speaker notes · Original schematic diagrams

Huawei Ascend: chips and complete systems

Accelerators, software, racks and networks require coordinated design.



Complementary layers of the Ascend stack, rather than a manufacturing flow.

Sources in speaker notes · Original schematic diagrams

Huawei announced roadmap

Distinguish corporate plans from verified availability.



Future announcements are roadmaps rather than certified availability or performance.

Dates and specifications reflect Huawei announcements in 2025–2026 and depend on execution.

Sources in speaker notes · Original schematic diagrams

SMIC, lithography and process yield

Industrial capability depends on equipment, process recipes, yield and scale.

01

PROCESS

DUV multi-patterning and additional processing complexity.

02

EQUIPMENT

Dependencies on advanced lithography and metrology.

03

YIELD

Process maturity and good chips per wafer.

04

SCALE

Learning cycles, energy, materials and capacity.

Production levels require evidence on yield and costs.

Sources in speaker notes · Original schematic diagrams

Memory and complementary suppliers

DRAM, NAND and HBM involve different designs and supply chains.

A

CXMT / DRAM

Volatile memory and DRAM manufacturing. HBM also requires specialised stacking and integration.

B

YMTC / NAND

Non-volatile storage. NAND cannot replace HBM in AI accelerators.

NAND manufacturing capacity does not establish HBM capacity.

Sources in speaker notes · Original schematic diagrams

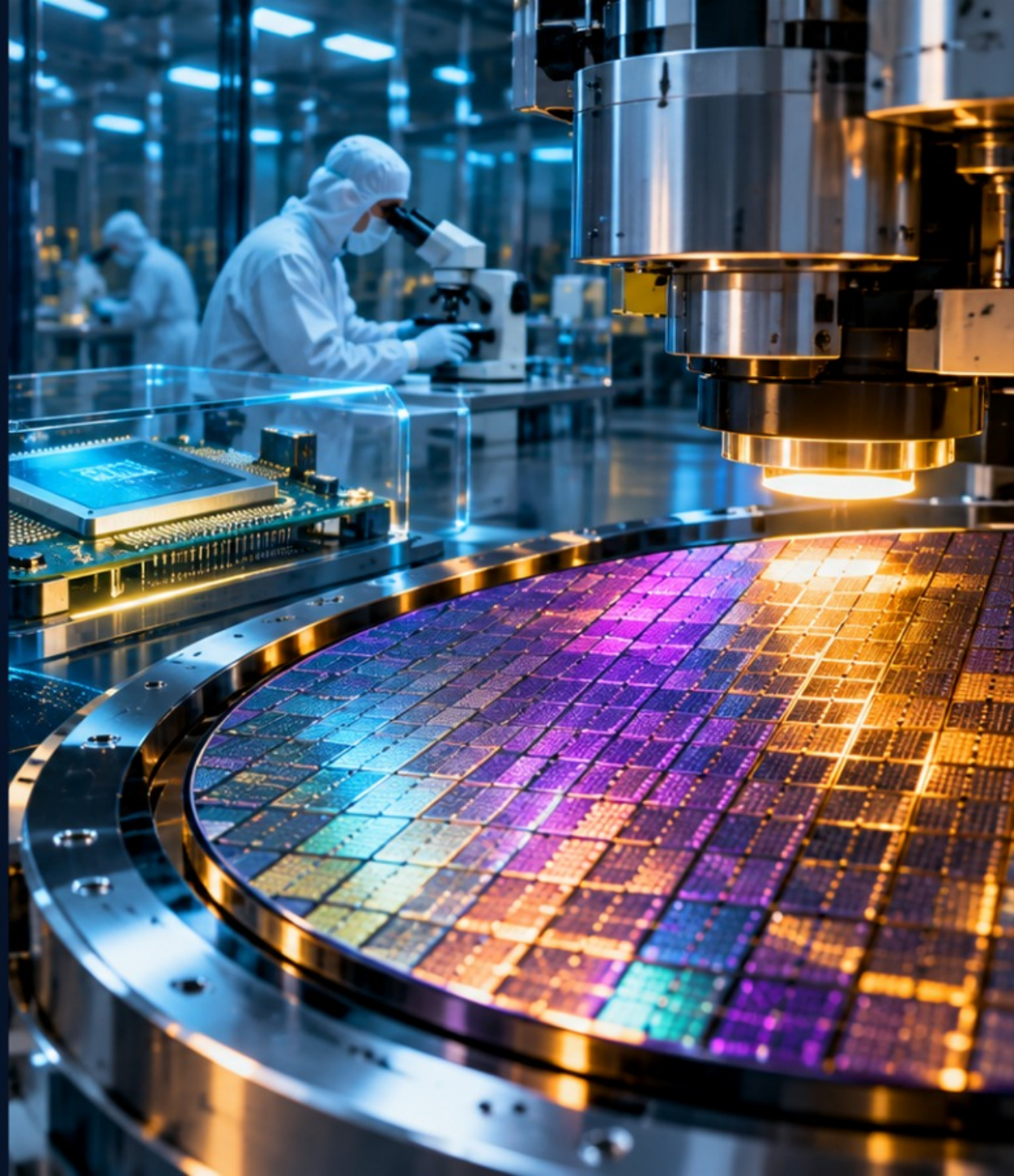
BLOCK 05 / 07

From accelerators to data centres

Distributed computing, networks and energy

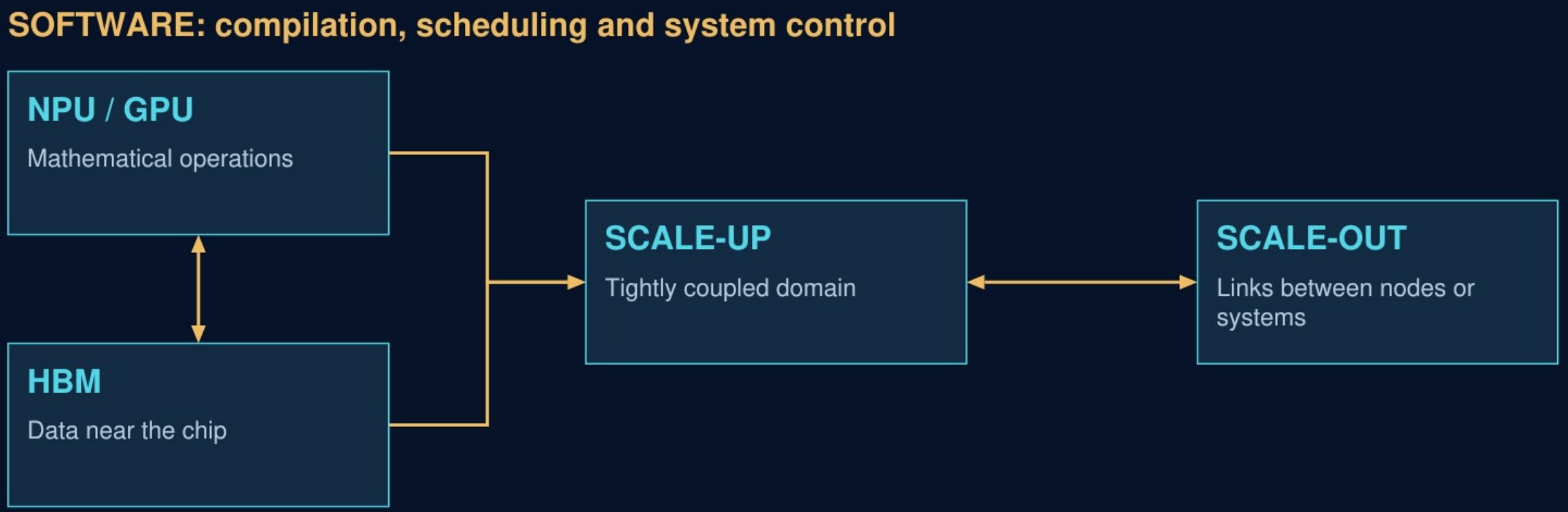
What does a cluster gain from adding accelerators?

SERGIO GEVATSCHNAIDER



Server architecture

Systems need host CPUs, memory, interconnects and cooling.



Functional schematic. Scale-up and scale-out boundaries depend on system architecture.

Sources in speaker notes · Original schematic diagrams

Communication overhead

Scaling can increase traffic and reduce efficiency.

More accelerators ≠ unlimited linear scaling



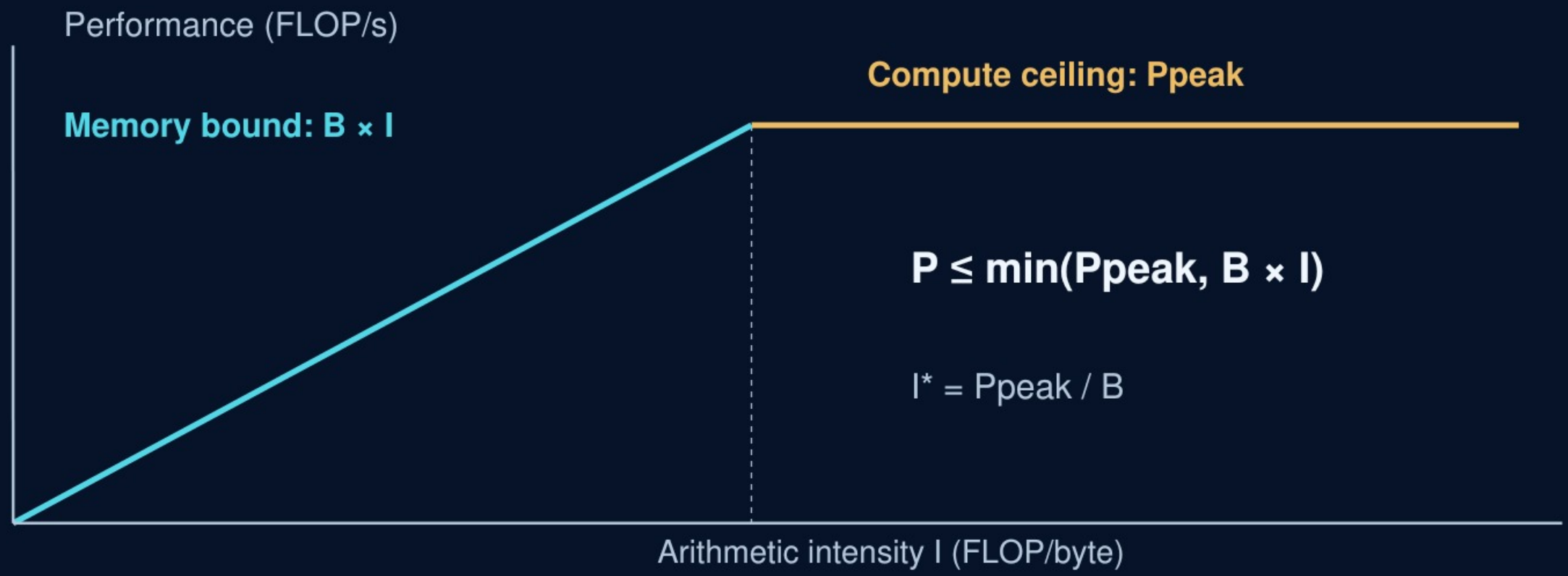
Communication, memory, fragmentation and synchronisation reduce efficiency.

Teaching schematic of communication overhead, without representing a commercial product.

Sources in speaker notes · Original schematic diagrams

Roofline: compute and memory

Arithmetic intensity determines which resource limits a workload.

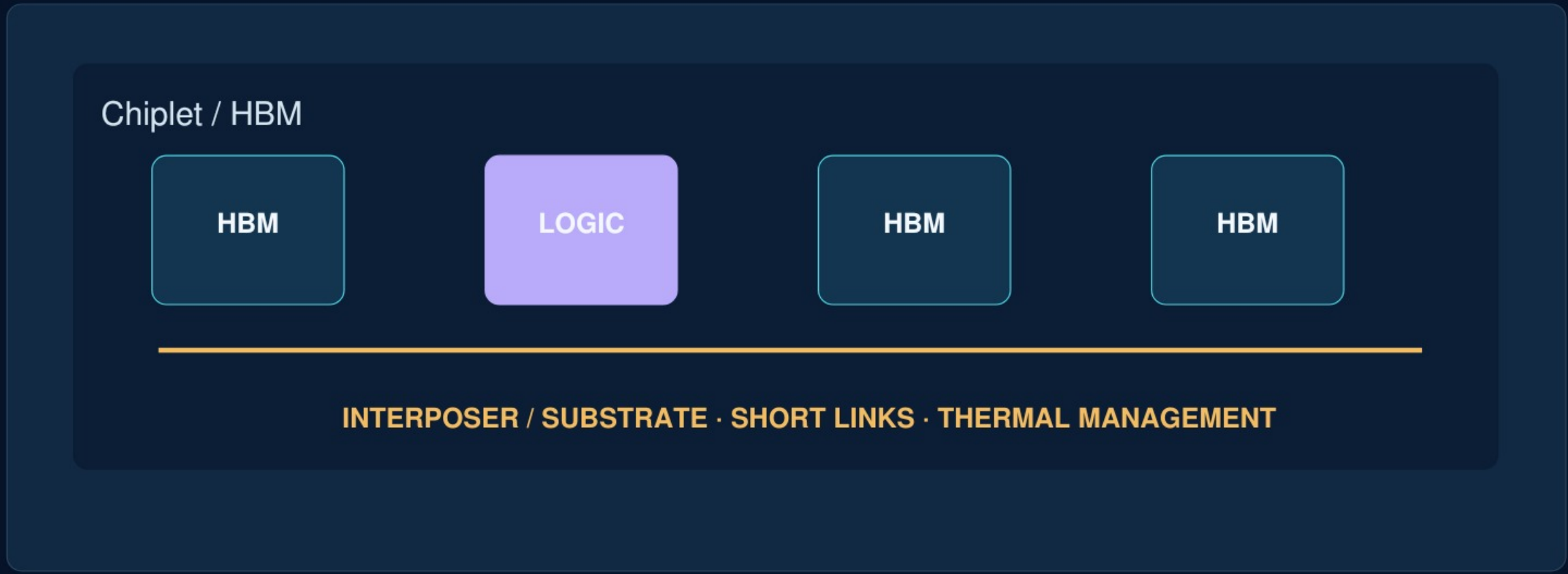


Conceptual bounds meet at I*. Observed performance may lie below the bound.

Sources in speaker notes · Original schematic diagrams

HBM and packaging

High-speed memory needs physical proximity and integration.



More compute without matching bandwidth can leave resources idle.

Sources in speaker notes · Original schematic diagrams

Cluster energy and operations

Measure useful tasks per unit of energy, alongside peak compute.

01

IT POWER

Accelerators, switches, CPUs, memory and storage.

02

PUE

Total data-centre power divided by IT power.

03

COOLING

Thermal limits and cooling-system choices.

04

UTILISATION

Scheduling, congestion and idle periods.

Electricity is a central economic input in AI expansion.

Sources in speaker notes · Original schematic diagrams

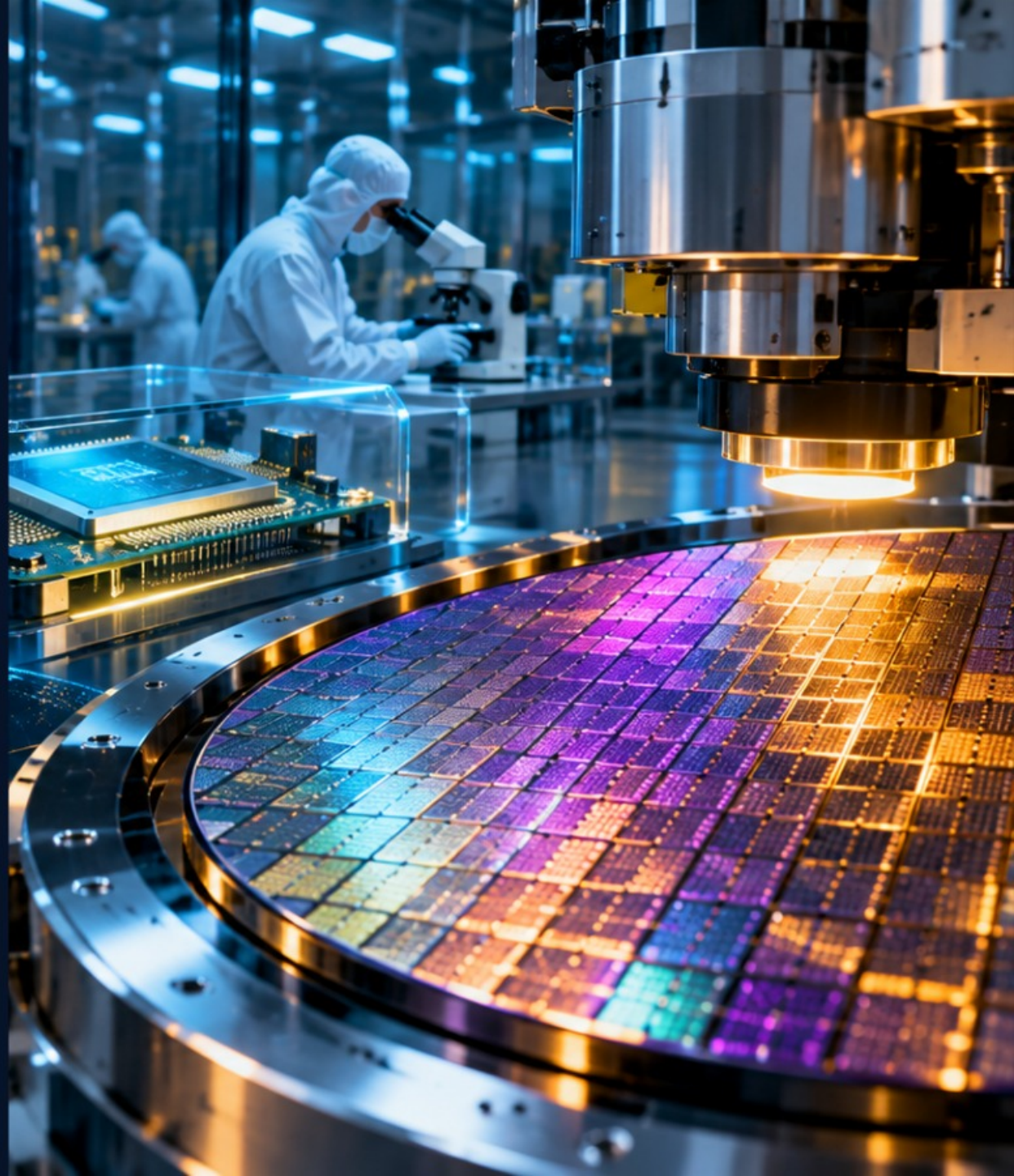
BLOCK 06 / 07

Geoeconomics and regulation

Interdependence and export controls

What does competition under changing restrictions involve?

SERGIO GEVATSCHNAIDER



Policy instruments

Trade rules, subsidies, research and standards interact.

01

EXPORTS

Licences, controls and destination/end-user rules.

02

CAPITAL

Capacity incentives and investment policies.

03

TECHNOLOGY

Intellectual property, talent and standards.

04

MARKETS

Customers, software, ecosystems and adoption at scale.

Export controls do not always imply an absolute prohibition.

Sources in speaker notes · Original schematic diagrams

US licensing policy: January 2026

In January 2026, BIS introduced case-by-case review for certain chips.

A

RULE

BIS changed licence review for the H200, MI325X and comparable products, subject to conditions.

B

INTERPRETATION

Approval and unrestricted access do not follow automatically. Product, recipient and licence conditions matter.

Historical regulatory example: BIS and Federal Register, January 2026.

Sources in speaker notes · Original schematic diagrams

Bottlenecks and substitution

A restriction can redirect investment towards another stage.



LIMITING STAGE → EUV

Fictional vulnerability indices, without measurements of China.

Sources in speaker notes · Original schematic diagrams

Three possible economic responses

Investment is one of several routes to autonomy.

01

SUBSTITUTION

Develop local suppliers, technology and capabilities.

02

ARCHITECTURE

Redesign clusters, algorithms and the division of work.

03

INTERDEPENDENCE

Preserve feasible and efficient trade flows.

Complementary strategies involve different costs and time horizons.

Sources in speaker notes · Original schematic diagrams

Implications for Latin America

Supplier structures affect industry, cloud services and digital policy.

01

PRICES

Technology competition and infrastructure costs.

02

ACCESS

Services, equipment and supplier diversity.

03

RULES

Data protection, compatibility and cybersecurity.

04

CAPABILITIES

Talent, applied research and productivity.

Opportunities depend on the sector and national public policy.

Sources in speaker notes · Original schematic diagrams

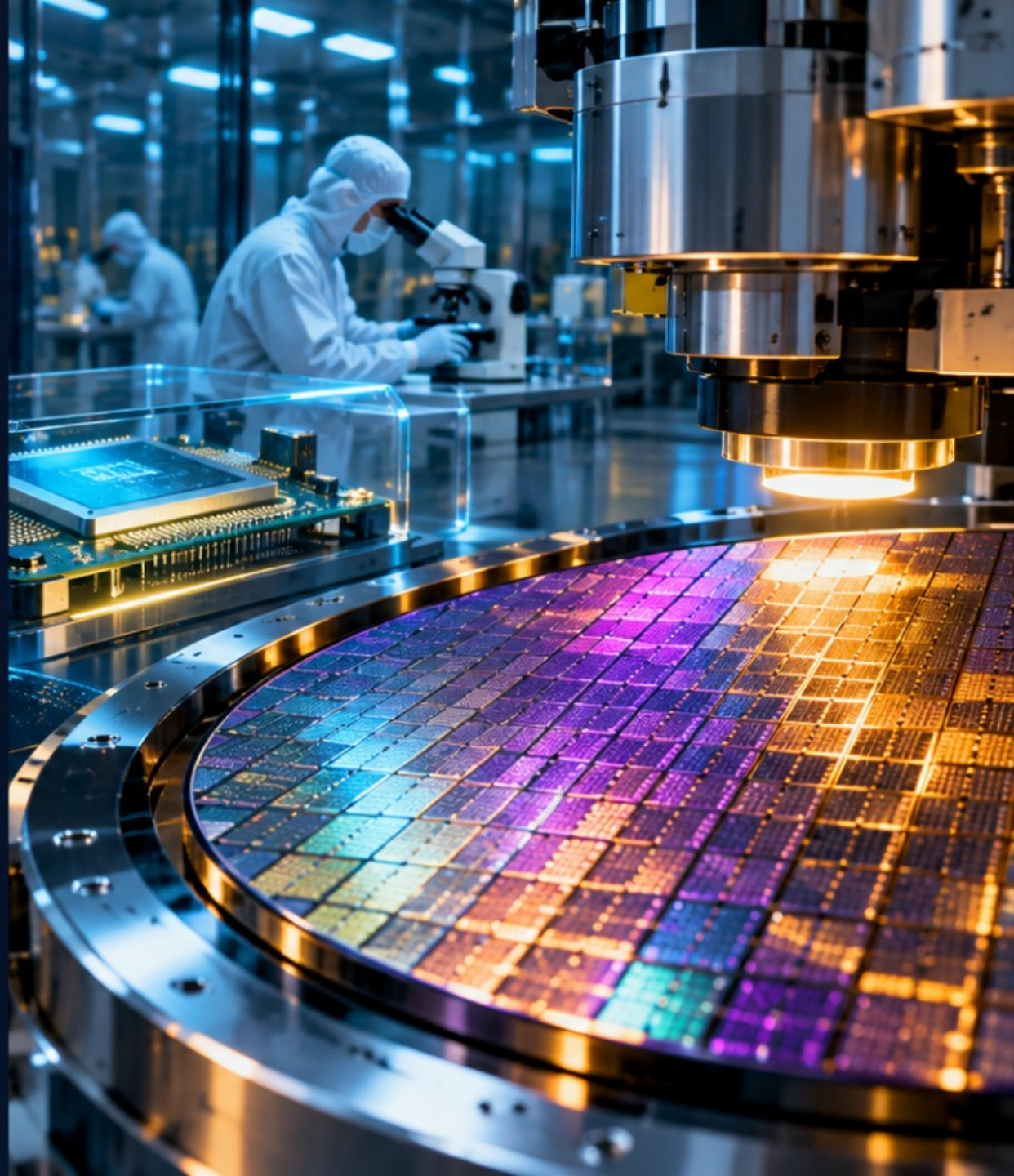
BLOCK 07 / 07

Scenarios and technological prospects

Industrial strategy and outcomes

Which questions should guide an assessment of future capabilities?

SERGIO GEVATSCHNAIDER



Four levers of technological capacity

The HTML laboratory explores production constraints.

01

FAB + YIELD

More functional dies with the same wafer capacity.

02

HBM

Memory supply compatible with the accelerators.

03

PACKAGING

Integration and testing at scale.

04

INTERCONNECTS

Convert delivered chips into useful compute.

Final production is limited by several complementary stages.

Sources in speaker notes · Original schematic diagrams

Laboratory scenarios

Five experiments compare investment strategies and shocks.

1	**Reference**	Moderate expansion in capacity and demand.
2	**Memory constraint**	HBM limits completed chips.
3	**Packaging expansion**	More assembly capacity may not increase deliveries.
4	**Yield improvement**	More good dies per wafer.
5	**Efficient networks**	More effective compute without manufacturing more chips.

All parameters are teaching assumptions rather than industrial estimates.

Sources in speaker notes · Original schematic diagrams

Interpreting simulation results

Simulation explores mechanisms. Forecasting requires calibration.

A

SUPPORTED INFERENCE

Constraints, sensitivity, tradeoffs and relative effects under explicit assumptions.

B

UNSUPPORTED INFERENCE

Company sales, yield or actual capacity without external evidence.

Separate conceptual sensitivity from verified empirical claims.

Sources in speaker notes · Original schematic diagrams

A decision framework for 2030

Assess technology choices by cost, robustness and learning.

01

SHORT TERM

Optimise software, utilisation, inventories and workload mix.

02

MEDIUM TERM

Integrate HBM, packaging, equipment and better yields.

03

LONG TERM

Develop process technology and industrial ecosystems.

04

CRITERION

Cost per useful task, safety, quality and resilience.

This analytical framework does not forecast commercial outcomes.

Sources in speaker notes · Original schematic diagrams

Five findings for analysing China

Competition involves complete ecosystems.

1	**Design and fabrication**	Manufacturing alone does not deliver a complete accelerator.
2	**Memory matters**	HBM can limit effective capacity.
3	**Networks matter**	Clusters incur scaling losses.
4	**Policy changes**	Regulations and strategies evolve.
5	**System-level evaluation**	Measure quality, cost, energy and supply.

Assess technological leadership across layers rather than using a single figure.

Sources in speaker notes · Original schematic diagrams

Technological capacity across the supply chain

Five dimensions of autonomy and computing power

1	**Production capacity**	Wafers, yield, equipment and scale.
2	**Integration**	HBM, packaging, testing and cooling.
3	**AI systems**	Software, networks and utilisation.
4	**Political economy**	Investment, licensing and end markets.
5	**Evidence**	Distinguish measured data, announcements and simulations.

Documentation and further reading

Selected primary and technical sources

1	**ASML**	DUV/EUV principles, optics and lithography.
2	**WSTS / SIA**	Worldwide semiconductor market data for 2025.
3	**Huawei**	Ascend, SuperPoD and announced roadmaps.
4	**BIS / Federal Register**	US licensing policy in January 2026.
5	**Intel / SIA**	Fabrication, advanced packaging and materials supply chains.

Full source links appear in the speaker notes and audiovisual guide.